

Bachelor of Technology — Chemical Engineering with Data Analytics

Table 1: Chemical Engineering with Data Analytics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	56
	Discipline Electives	15
Electives	Free Electives	21
	MTP (4+4, equivalent as DE)	8
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 2: B. Tech Chemical Engineering with Data Analytics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CED-101	Introduction to Chemical Engineering		3
2	—	Chemical Engineering Thermodynamics		3
3	—	Fluid Mechanics	2-1-2-4	4
4	—	Numerical Methods	3-1-0-4	4
5	—	Chemical Reaction Engineering	2-1-2-4	4
6	—	Heat Transfer	2-1-2-4	4
7	—	Mass Transfer I	2-1-2-4	4
8	—	Mass Transfer II		3
9	—	Data Science for Chemical Systems		3
10	—	Process Control and Instrumentation	2-1-2-4	4
11	—	Chemical Reaction Engineering II		3
12	—	Transport Phenomenon		3
13	—	Digital Process Equipment Design		3
14	—	Chemical Process Technology		3
15	—	Process Engineering and Economics		3
16	—	Chemical Engineering Lab (Remaining Experiments)		2
17	—	Climate Technology and Sustainable Process Engineering		3
Total				56

Table 3: B. Tech Chemical Engineering with Data Analytics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
6		IC-182	History of Science and Technology	3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language I	3-0-0-3	
9		HS-342	German I	3-0-0-3	
10		IK-101	Sanskrit language-Level 1		
11	DC	CED-101	Introduction to Chemical Engineering		3
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Chemical Engineering with Data Analytics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket-I	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Chemical Engineering with Data Analytics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	—	Chemical Engineering Thermodynamics		3
4	DC	—	Fluid Mechanics	2-1-2-4	4
5	DC	—	Numerical Methods	3-1-0-4	4
6	IC Basket-II	IC-XXX	Material and Energy Balance		3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Chemical Engineering with Data Analytics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Chemical Reaction Engineering	2-1-2-4	4
2	DC	—	Heat Transfer	2-1-2-4	4
3	DC	—	Mass Transfer I	2-1-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Chemical Engineering with Data Analytics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Mass Transfer II		3
2	DC	—	Data Science for Chemical Systems		3
3	DC	—	Process Control and Instrumentation	2-1-2-4	4
4	DC	—	Chemical Reaction Engineering II		3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Chemical Engineering with Data Analytics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Transport Phenomenon		3
2	DC	—	Digital Process Equipment Design		3
3	DC	—	Chemical Process Technology		3
4	DC	—	Process Engineering and Economics		3
5	DC	—	Chemical Engineering Lab (Remaining Experiments)		2
6	DC	—	Climate Technology and Sustainable Process Engineering		3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Chemical Engineering with Data Analytics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	—	Undergraduate Research Project I	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Chemical Engineering with Data Analytics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	—	Undergraduate Research Project II	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 21 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*